

Blue Carbon Potential of Tidal Flats and Living Shorelines



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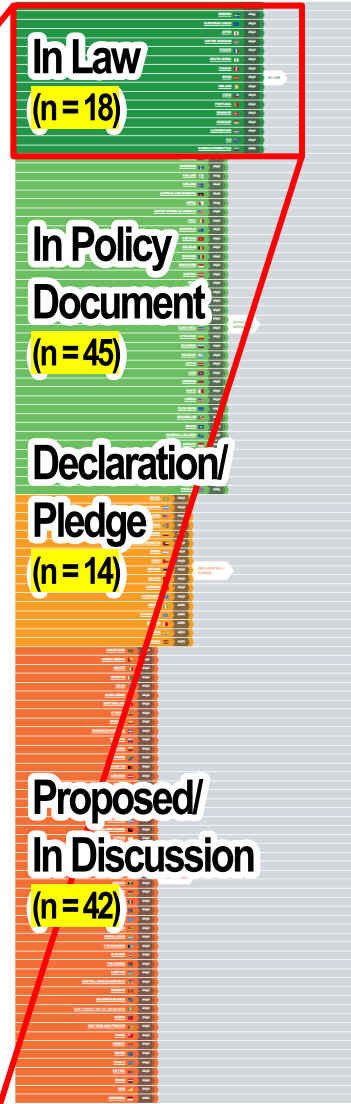
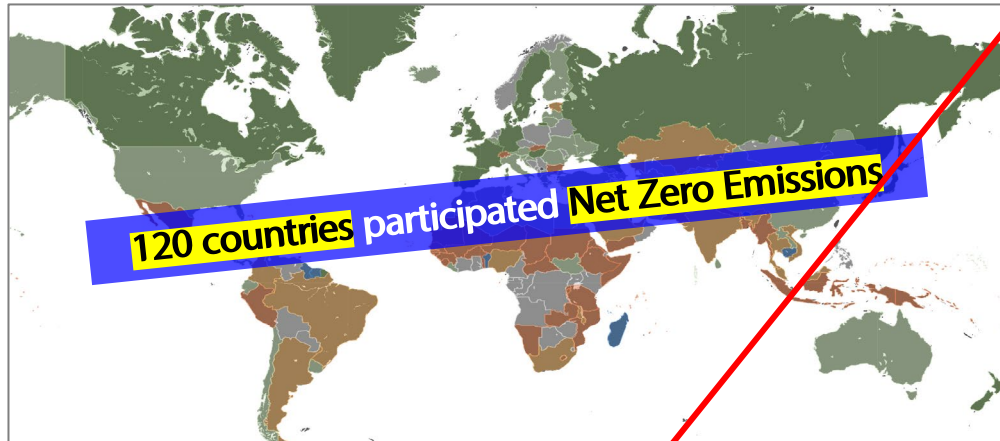


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 - 2. Korean Blue Carbon Science**
 - 3. Key Findings & Discussion**
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- 

1. Backgrounds

Net Zero Emissions Race(NDC & Net-Zero): K-MOF challenged Negative Emissions

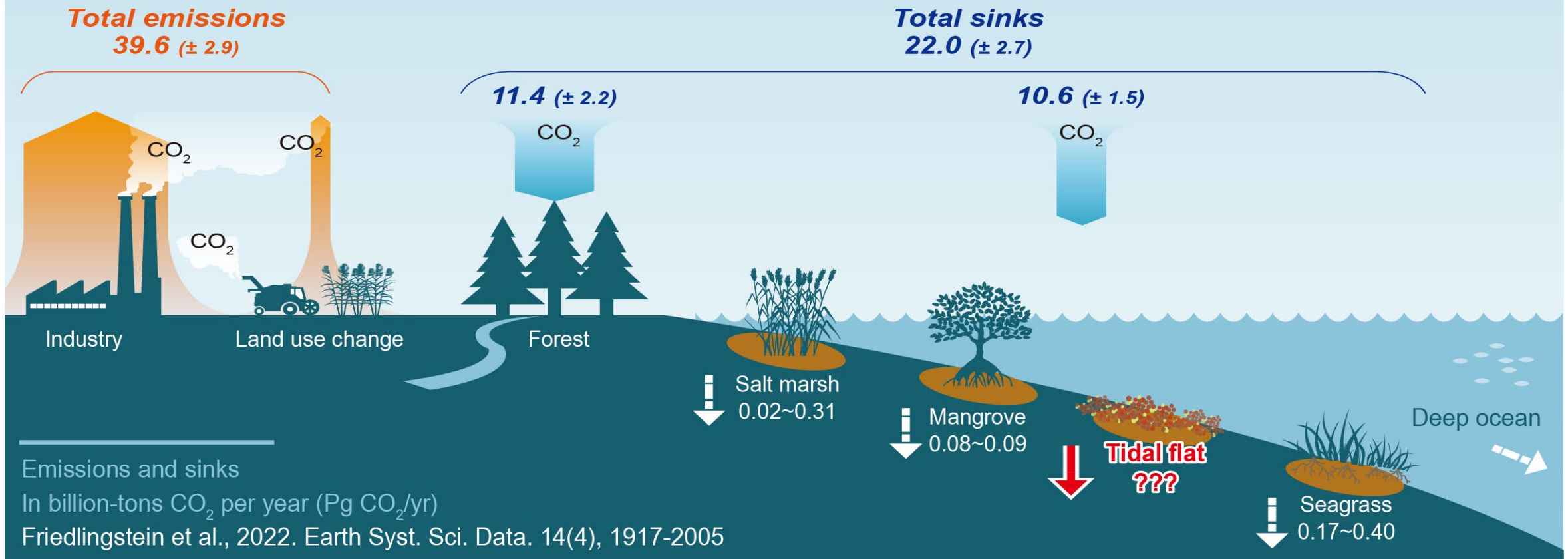


1. Backgrounds

The Importance of the Ocean in Response to Climate Change

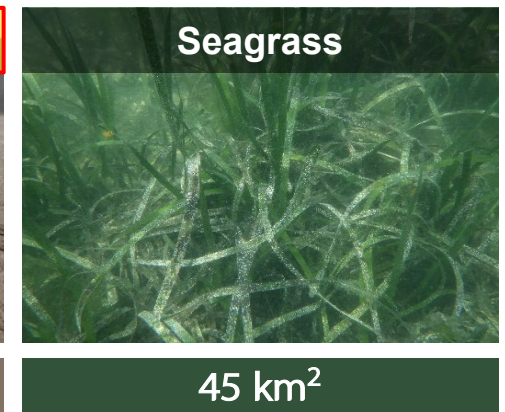
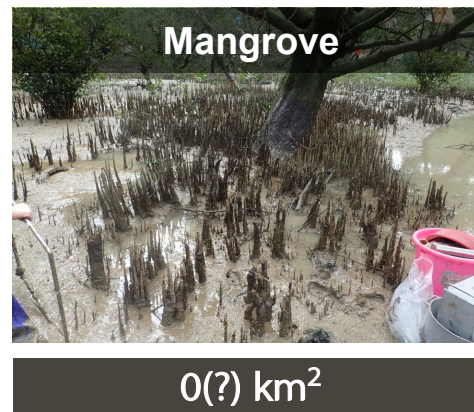
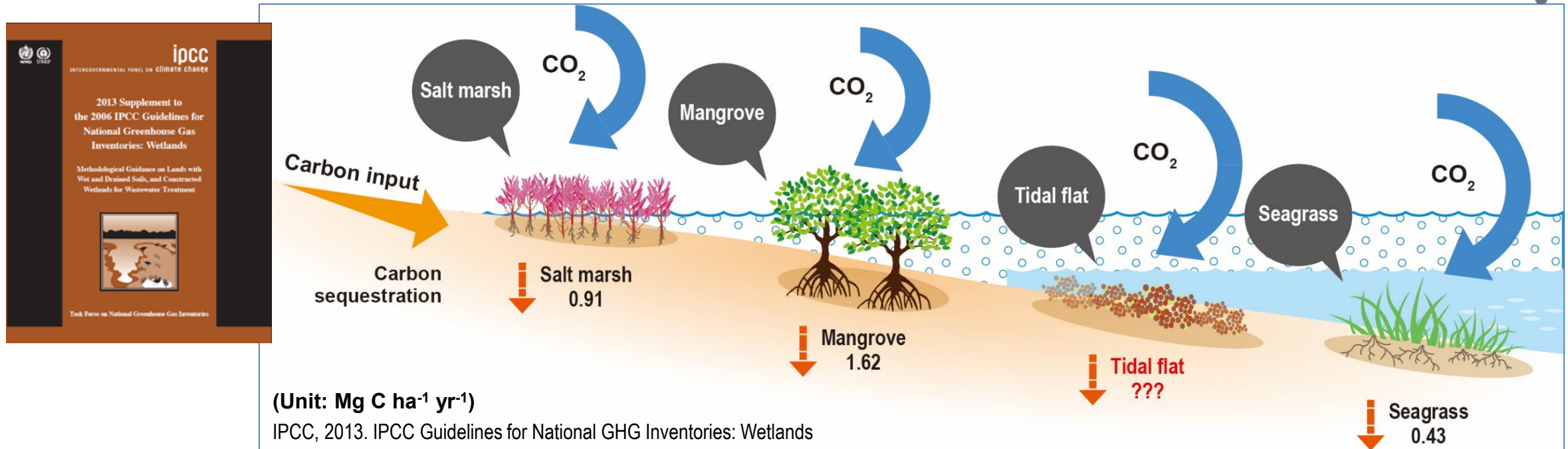
- ✓ Global carbon budget (2012-2021, 10yr average): CO₂ emissions vs. CO₂ Sinks
- ✓ CO₂ emissions: ~39.6 Pg yr⁻¹, Sinks: ~22.0 Pg yr⁻¹ (Green C 11.4 Pg yr⁻¹ ≈ Blue C 10.6 Pg yr⁻¹)

Global carbon budget 2012 - 2021



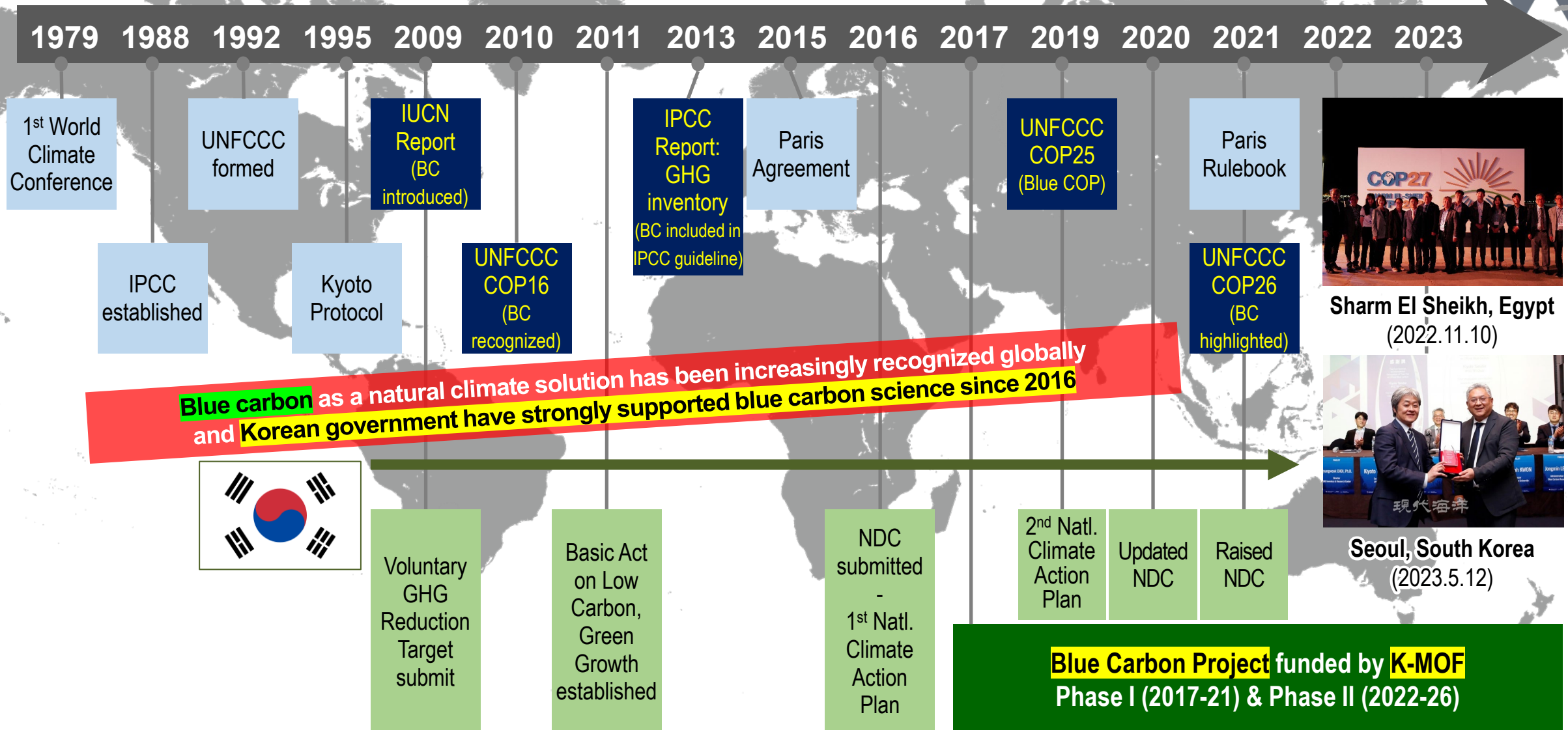
1. Backgrounds

We focused on the **“Tidal flat”**, as an Emerging Blue Carbon



1. Backgrounds

Blue Carbon Science in a Global Window of Climate Change



2. Korean Blue Carbon Science

Blue Carbon Terminology, **Carbon Stock = Carbon Absorption + Carbon Sequestration**

TERMINOLOGY

갯벌 깊이 약 40m (약 10만년 퇴적)
 갯벌 퇴적: 퇴적? continuous?
 10만 갯벌 깊이 (약 1mm/yr)
 해겟벨은 퇴적물의 10m 이상이다.

Carbon sink

저장 = 흡수 + 침적

Stock (Storage)
 Absorption
 = Biomass organic
 Carbon stock
 (through fixation
 process)
 * fixation
 = assimilation
 sequestration
 sedimentary
 organic carbon
 stock
 (through sedi-
 mentary
 burial)
 퇴적물
 분해
 탄소 저장량

- 갯벌에서 유기탄소가 퇴적층으로 침적되어 격리되었다.
- 산림은 대기 CO₂를 흡수하여 성장한다.
- CO₂ 저장이란 생물체 흡수와 퇴적물 내 침적을 총칭하여 일컫는 말이다.

난 탈해성
 생 분해성

유기탄소 저장량(현량)

유기탄소 침적률

블루카본 후보(갯벌)

탄소 흡수 (고정+분해)

탄소 침적 (고정)

탄소 침적

탄소 배출

유지 정도

퇴적물

분해

탄소 저장량

Refractory (물리·화학적 축적) (<-> SWS)

Recalcitrant (생물학적 축적)

Labile <-> Recalcitrant

Organic carbon stock (유기탄소 저장량)

Organic carbon sequestration rate (유기

Carbon capture (탄소포집)

Carbon absorption (탄소흡수)

Carbon sequestration (탄소침적)

Carbon burial (탄소침적)

Carbon emission (Respiration, Decomposition)

Bulk density (용적밀도) (= released) (탄소배출)

Sedimentation rate (퇴적률)

Decomposition (분해)

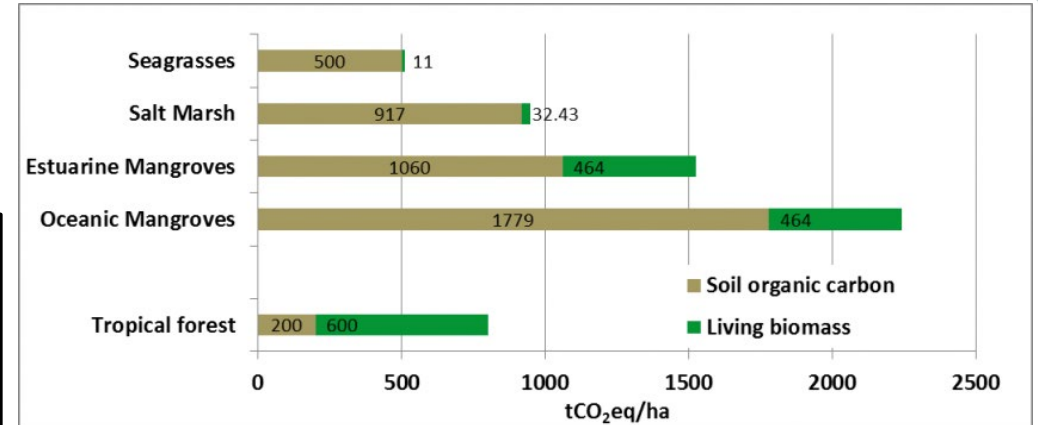
Carbon storage (탄소저장)

Carbon budget (탄소예산)

Carbon reduction (탄소감축(원)) (=탄소저감)

Carbon mitigation (탄소완화)

Carbon neutrality (탄소중립
 (= Net-zero carbon) = 탄소저감)



그린카본

CO₂ 고정
 (연간 약 2.4 Kg)

점유면적 12 m² > 식재면적 7 m²

식재비용

1,100,000원 >> 60,000원

관리비용

30,000원 (10년) > 없음

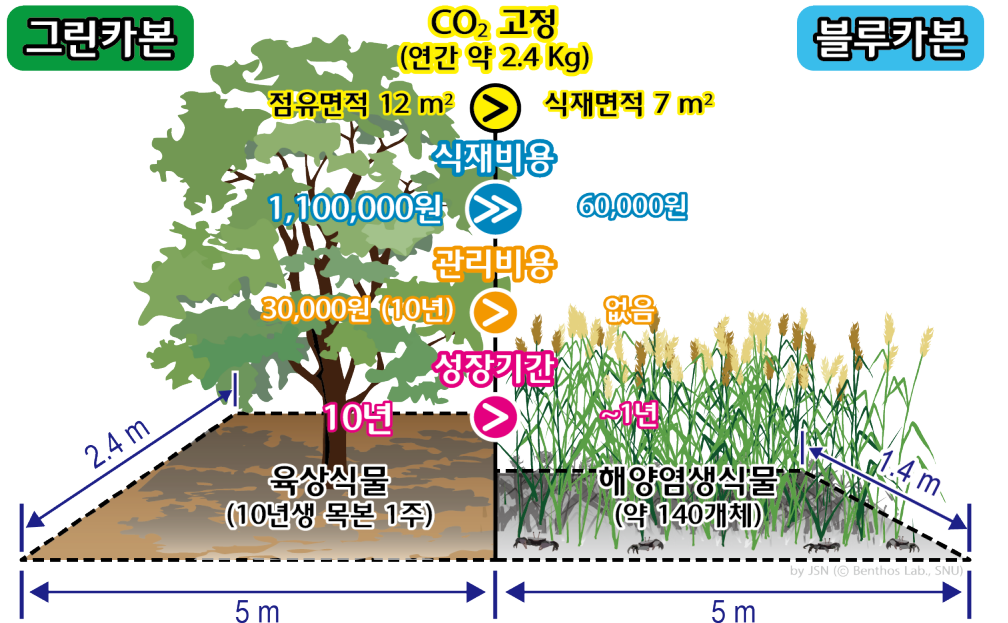
성장기간

10년 > ~1년

육상식물 (10년생 목본 1주)

해양염생식물 (약 140개체)

블루카본



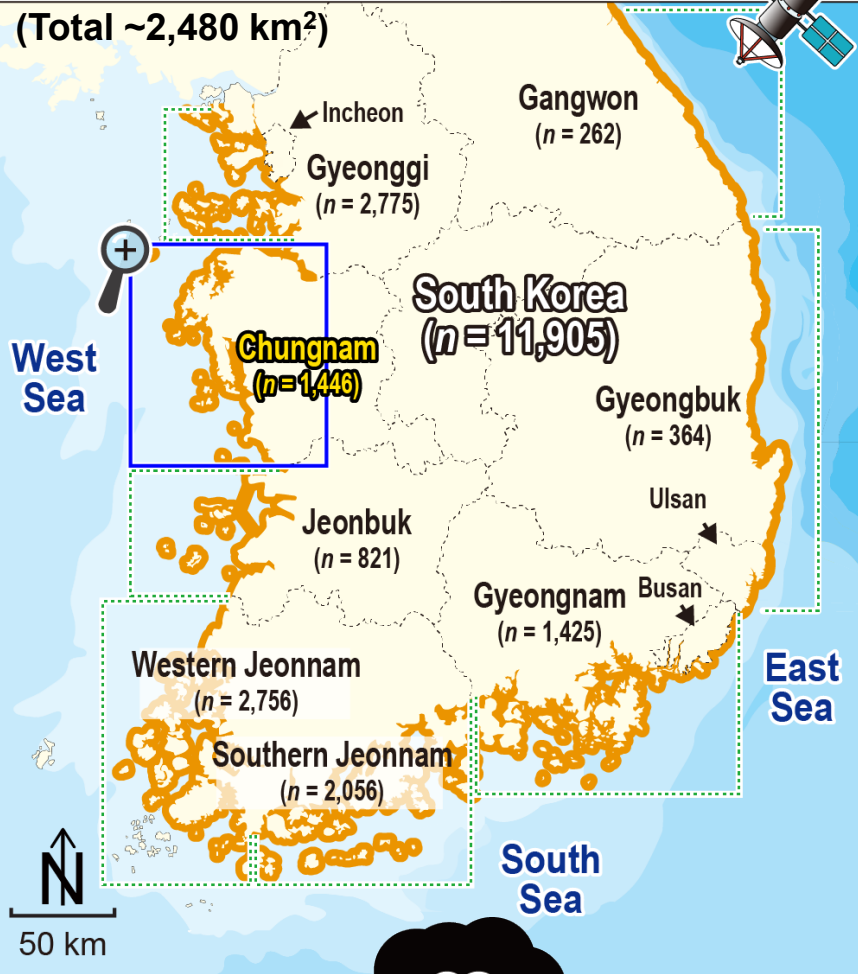
Phase-I (2017-2021): \$8 million dollars

16

2. Korean Blue Carbon Science

Phase-I (2017-2021)

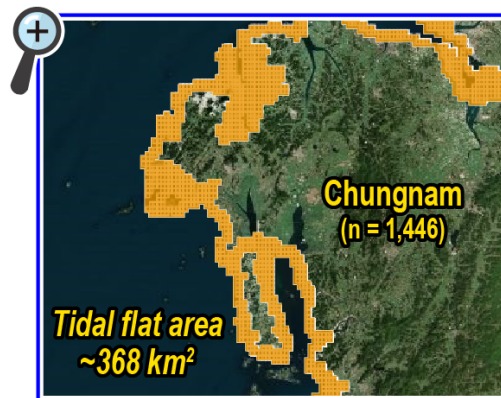
Intertidal area of South Korea



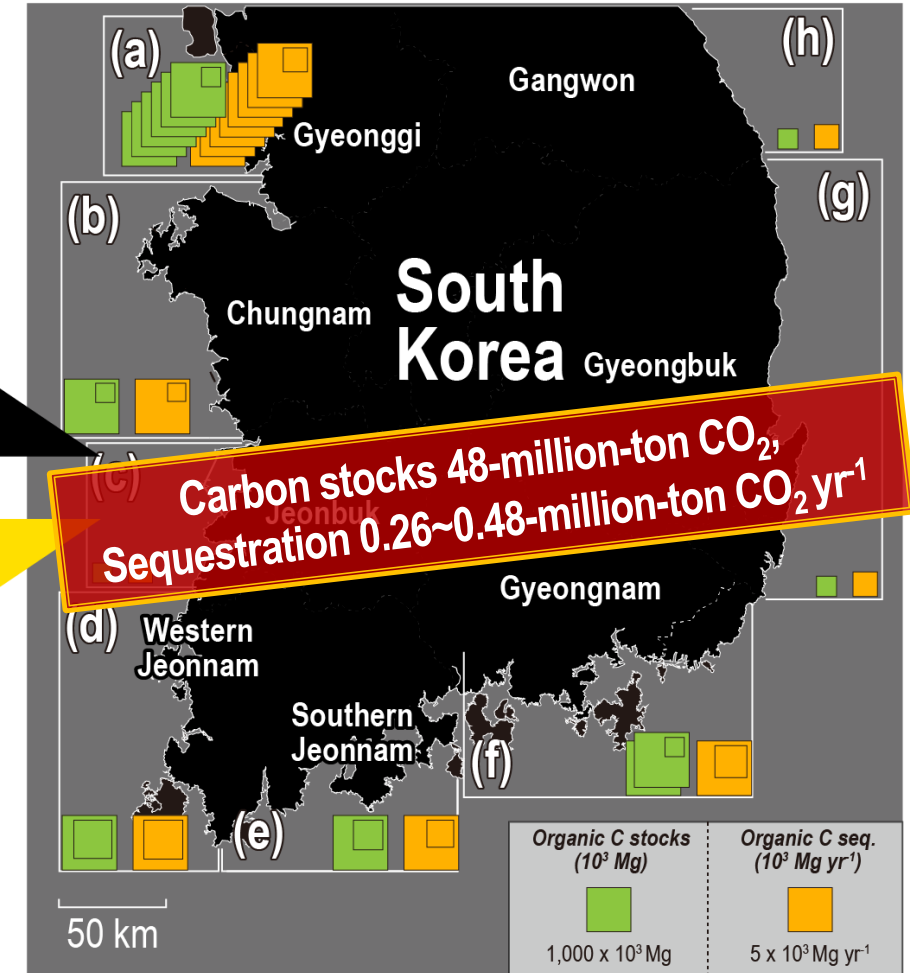
Field study (n = 303)

Sand Mixed Mud

Remote sensing (n = 11,905)



Lee et al., 2021



Tidal flats can absorb CO₂, which generates 110,000~200,000 cars a year!!

2. Korean Blue Carbon Science

Phase-II (2022-2026): \$40 million dollars

Blue Carbon Project: Phase-II

2021

R&D strategies for
climate change
Research planning
(SNU)

Living shoreline
Research planning
(SNU)

블루카본 기반 기후변화 적응형
해안조성기술개발 사업 기획연구
- 본 보고서 -

2021. 9. 9.

서울대학교
산학협력단

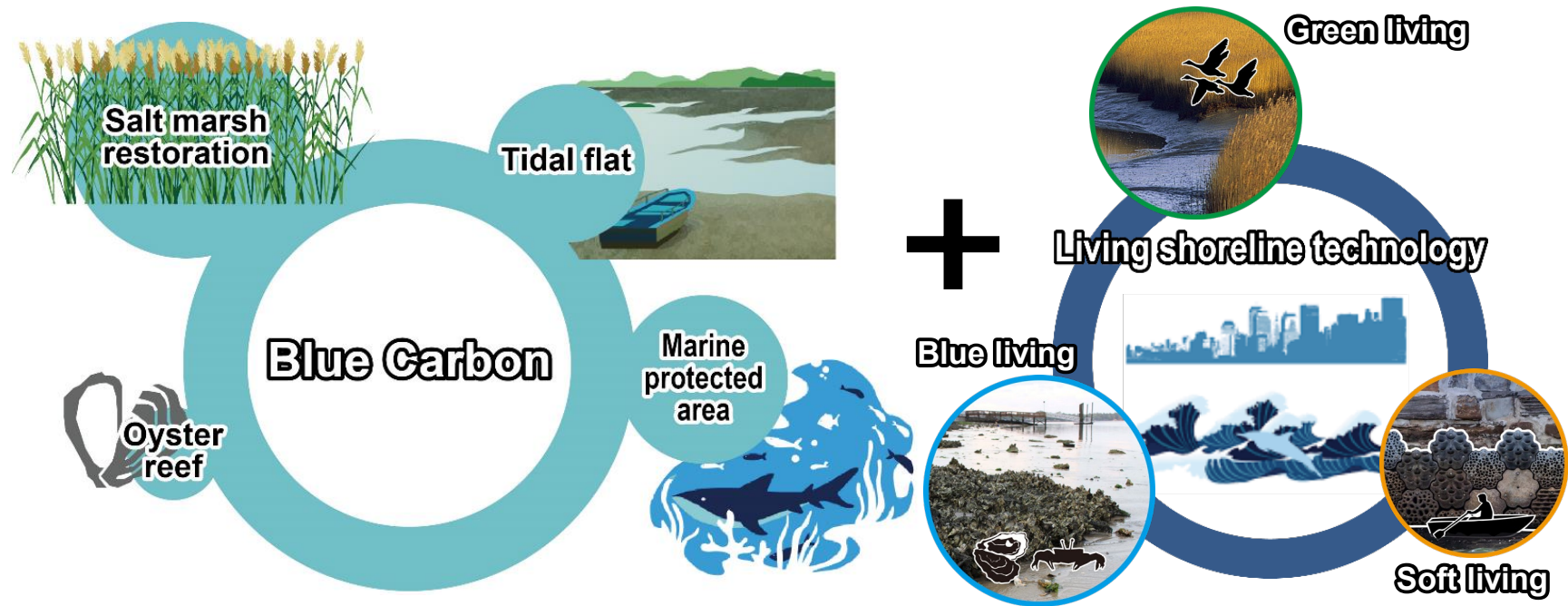
2022-26

Blue Carbon
Project (phase-II)
→ On going('22~'26)
(SNU)



Blue Carbon
Research Center

[Title] Development of living shoreline technology based on blue carbon science toward climate change adaptation



The project aims to support **net-zero by 2050** through
1) Enhancement & Excavation of **blue carbon**
2) Construction & management of **living shoreline** technology

2. Korean Blue Carbon Science

Phase-II (2022-2026): Objectives, contents, and expected outcomes

Research objectives

Aimed to develop techniques on enhancement of “**blue carbon resources**” and apply in situ along the coasts of Korea using the “**living shoreline techniques**” towards “**carbon neutrality**”

Major expected outcomes

- **Blue Carbon Map of Tidal flats**
 - Excavation of new carbon reduction resources
 - New Blue Carbon absorption/emission factors
 - Inclusion of information about tidal flat area (>80%)
- **International network operation (IPCC, COP, etc.)**
- **Development of Living shoreline technology**
 - Test-bed demonstration & Guidelines suggestion
 - Technical guidelines for Living shoreline
 - Announcement of coastal management

Three major subjects & research contents

1. Enhancement & Excavation of Blue Carbon

Subject 1

- 1.1. Identification of carbon cycle in tidal flats
- 1.2. Excavation of new carbon reduction resources
- 1.3. Development of remote sensing technology
- 1.4. Construction of **Blue Carbon database**
- 1.5. Reinforcement of international network

2. Construction of Living Shoreline

Subject 2

- 2.1. Development of Living shoreline technology
- 2.2. Analysis of Living shoreline effects
- 2.3. **Test-bed operation** & Technology demonstration

3. Management of Blue Carbon & Living Shoreline

Subject 3

- 3.1. Enactment of Law & System
- 3.2. Reinforcement of business in local government

3. Key Findings & Discussion

Criteria for Inclusion as Actionable Blue Carbon Ecosystems

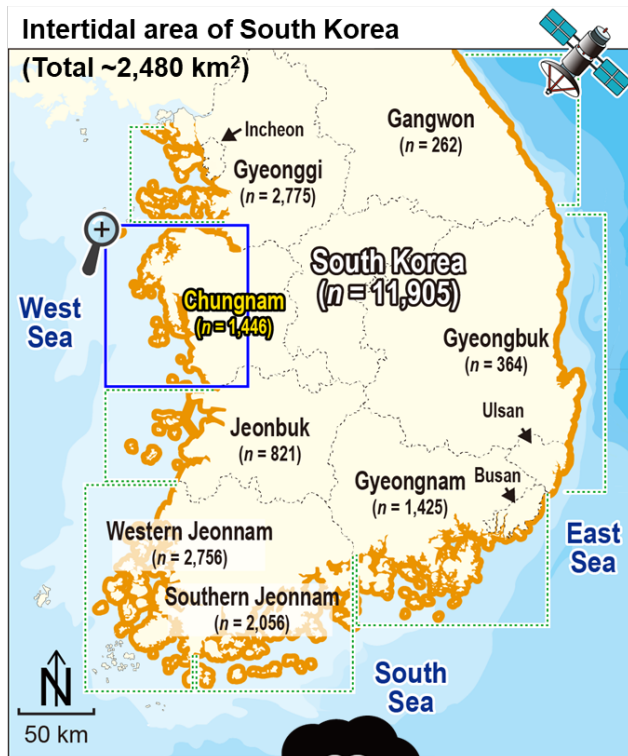
		Blue Carbon Criteria (Adapted from Lovelock & Duarte, 2019; Conservation International, 2021)					
Category	Habitat type	Large scale of GHG removals or emissions	Long-term storage of fixed CO ₂	Carbon loss by anthropogenic impacts	Practical management to reduce emission	Included in IPCC GHG accounting guidelines	Alignment with other policies
Actionable Blue Carbon Ecosystems for Mitigation	Mangrove	YES	YES	YES	YES	YES	YES
	Salt marsh	YES	YES	YES	YES	YES	YES
	Seagrass	YES	YES	YES	YES	YES	YES
Emerging Blue Carbon Ecosystems	Tidal flat	?	?	YES	?	NO	YES
	Benthic sediment	?	YES	YES	?	NO	?
	Macroalgae	YES	YES	YES	YES	NO	YES
Other Ocean Ecosystems (Not actionable)	Coral reef	NO	NO	NO	NO	NO	YES
	Oyster reef	NO	?	NO	NO	NO	YES
	Phytoplankton	YES	?	?	NO	NO	NO
	Fish	NO	NO	NO	NO	NO	YES

3. Key Findings & Discussion

BC criteria for **Tidal flats**: Large scale of GHG removals/emissions

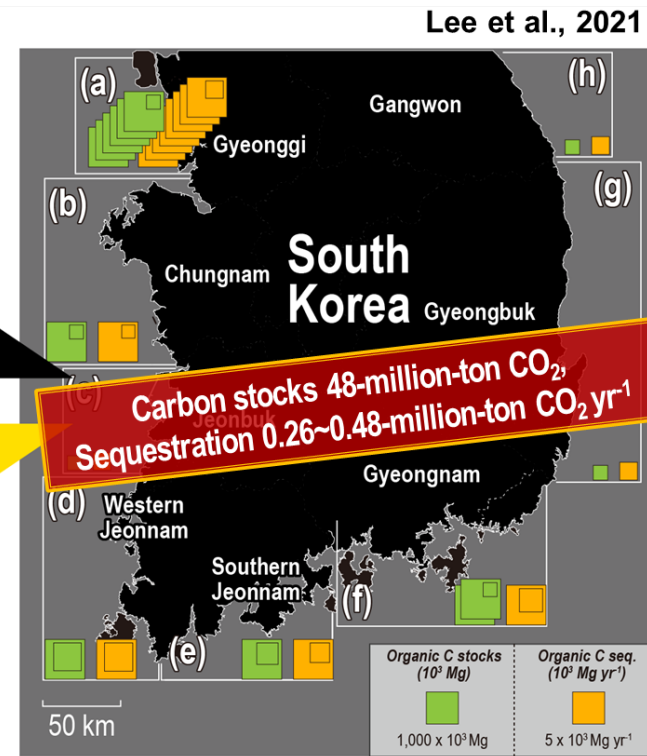
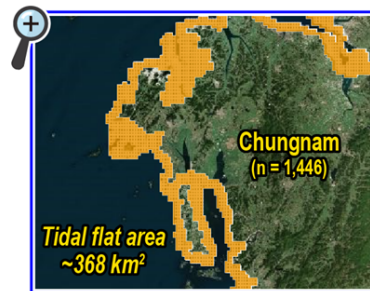
Category	Habitat type
Emerging Blue Carbon Ecosystems	Tidal flat

Blue Carbon Criteria (Adapted from Lovelock & Duarte, 2019; Pidgeon et al., 2021)					
Large scale of GHG removals or emissions	Long-term storage of fixed CO ₂	Carbon loss by anthropogenic impacts	Practical management to reduce emission	Included in IPCC GHG accounting guidelines	Alignment with other policies
? → YES	?	YES	?	NO	YES



Field study (n = 303)

Sand Mixed Mud
Remote sensing (n = 11,905)



연안습지 탄소흡수량 '블루카본' 국가 온실가스 통계 첫 반영 | 연합뉴스

(서울=연합뉴스) 차민지 기자 = 해양수산부는 염생 식물이 서식하는 연안습지의 탄소 흡수량(블루카본)을 산정해 올해부터 국가 온실가스 통계에 반영...

<https://www.yna.co.kr>

분야	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999
연안습지	-6	-10	-13	-14	-14	-7	-8	-8	-6	-3
	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
	-5	-5	-7	-10	-12	-5	-5	-6	-7	-7
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
	-7	-7	-9	-8	-8	-11	-9	-9	-11	-11
	2020	2019년 대비								
	-11	2.2%								



Tidal flats can absorb CO₂, which generates 110,000~200,000 cars a year!!

3. Key Findings & Discussion

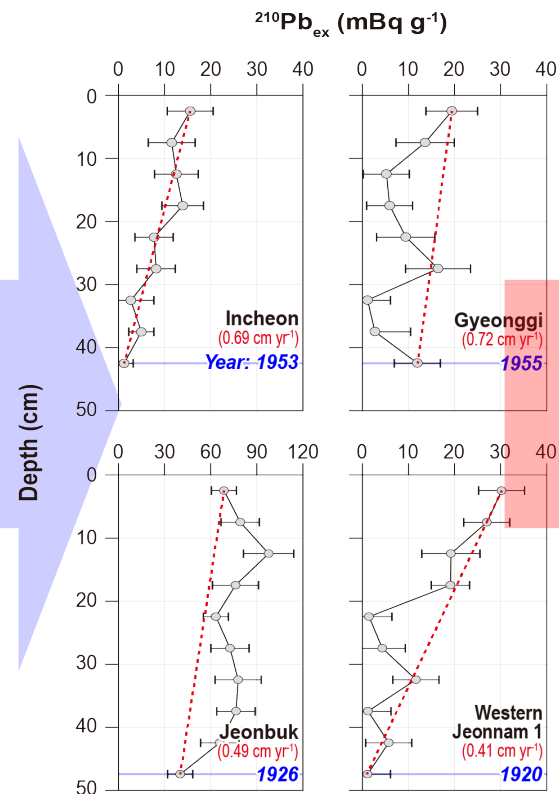
BC criteria for **Tidal flats**: Long-term storage of fixed CO₂

Blue Carbon Criteria (Adapted from Lovelock & Duarte, 2019; Pidgeon et al., 2021)

Category	Habitat type	Large scale of GHG removals or emissions	Long-term storage of fixed CO ₂	Carbon loss by anthropogenic impacts	Practical management to reduce emission	Included in IPCC GHG accounting guidelines	Alignment with other policies
Emerging Blue Carbon Ecosystems	Tidal flat	? → YES	? → YES	YES	?	NO	YES



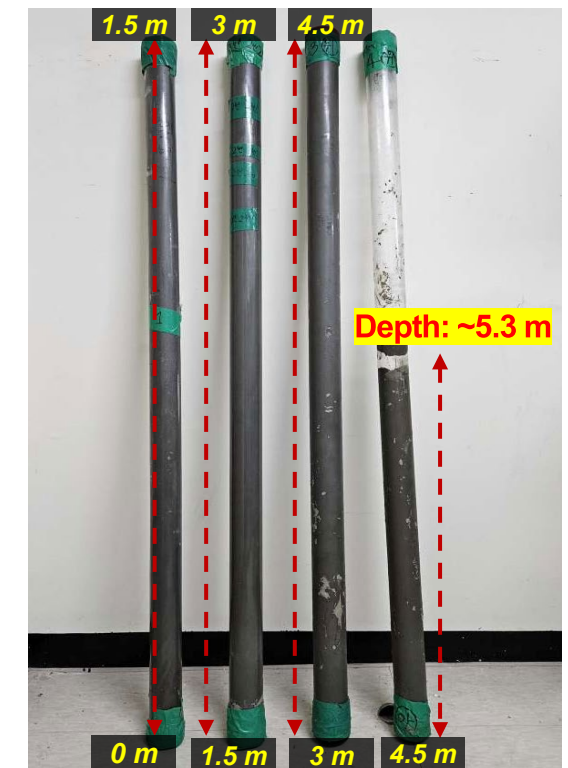
Current sediment coring
(max depth: ~1 m)



Lee et al, 2021



Upcoming sediment coring
(max depth: ~6 m)



BC criteria for Tidal flats: Practical management to reduce emission

The figure consists of a map of South Korea and an inset line graph. The map displays various marine protected areas, color-coded according to the legend: green for Wetland protected area, blue for Marine ecosystem protection area, dark blue for Marine species protection area, purple for Marine landscape protected areas, and orange for 'Getbol, Korean Tidal Flat, UNESCO World Natural Heritage. The map includes labels for numerous locations, such as Ungjinjangbongdo Tidal flat, Songdo Island Tidal flat, Siheung Tidal flat, Deabudo Tidal flat, near Daeijakdo Island, Hwaseong Maehyang-ri Tidal Flat, Garorim Bay, Sinduri Dune, Boryeong Sohwang Dune, Seocheon Tidal flat, Buanjulpo Bay Tidal flat, Gochang Tidal flat, Muan Tidal flat, Jungdo Tidal flat, 'Getbol, Korean Tidal Flat, UNESCO World Natural Heritage, Jindo Tidal flat, near Gageodo Island, near Munseom Island, near Tokkiseom Island, near Jodo Island, near Ulleungdo Island, Near Ulljin, Near Po, Near Goseong Haimyeon, near Namhyeongjesseom Island, near Namuseum Island, near Oryukdo Island, Near Tongyeong Seonchon Village, Masanmanbongam Tidal flat, Suncheon Bay Tidal flat, Goheung Tidal Flat, Boseongbeolgyo Tidal flat, near Sohwado Island, near Cheongsando Island, and near Chulado Island.

The inset line graph, titled "Marine protected area", shows the "Area (km²)" on the y-axis (ranging from 0 to 2000) against the year on the x-axis (ranging from 2001 to 2021). The graph shows a steady increase in the area of marine protected areas over time, with a significant jump around 2017, reaching approximately 1800 km² by 2021.

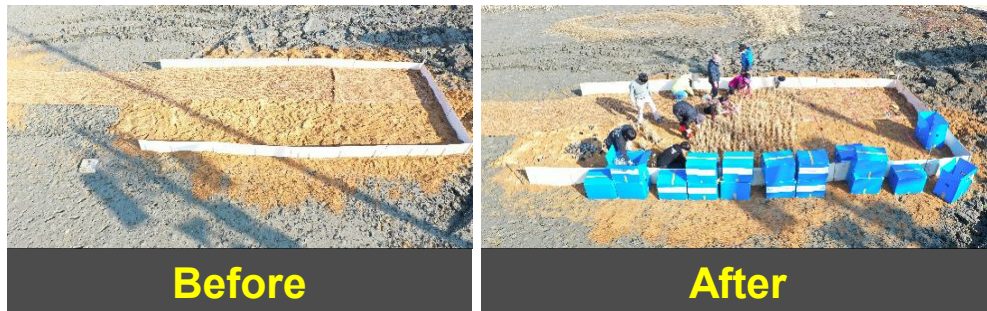


3. Key Findings & Discussion

BC criteria for **Tidal flats**: Practical management to carbon reduction

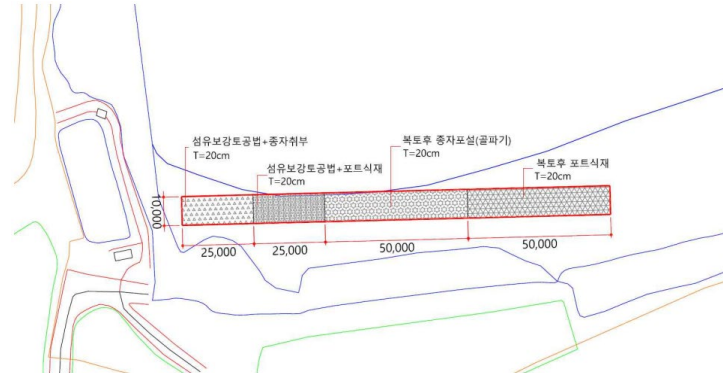
① **Green Living** Test-bed 1

(Location: **Daebu**; Monitoring: Oct. 2022 ~ Jul. 2023, 10 month)



② **Green Living** Test-bed 2

(Location: **Seocheon**; Monitoring: Jun. 2023 ~ Dec. 2026, 43 month)

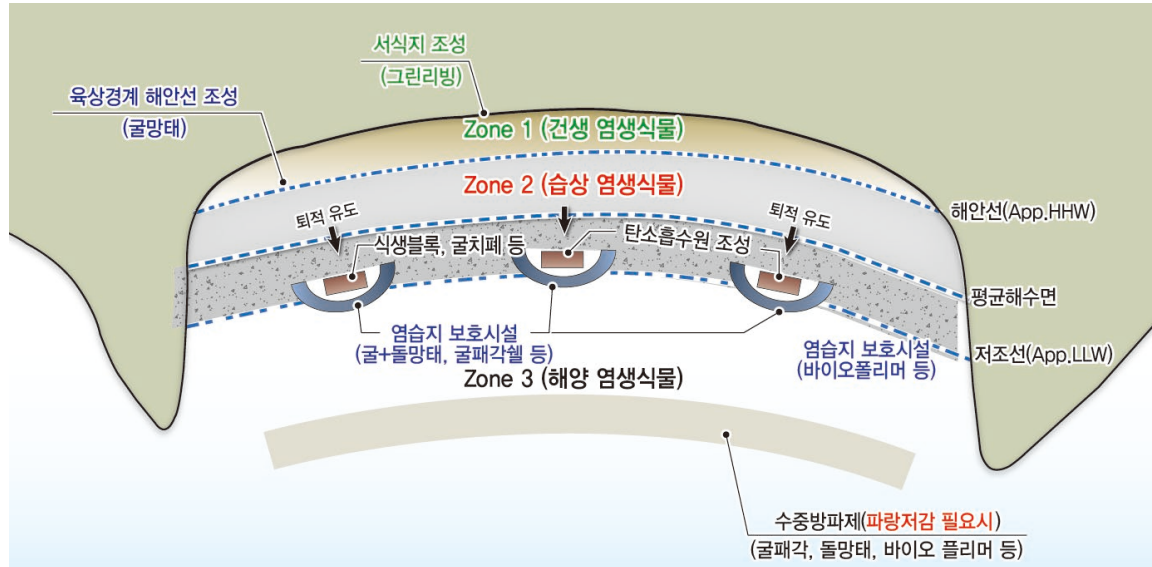


3. Key Findings & Discussion

BC criteria for **Tidal flats**: Practical management to carbon reduction

③ **Blue Living** Test-bed

(Location: **Gangjin**; Monitoring: Oct. 2023 ~ Dec. 2026, 39 month)

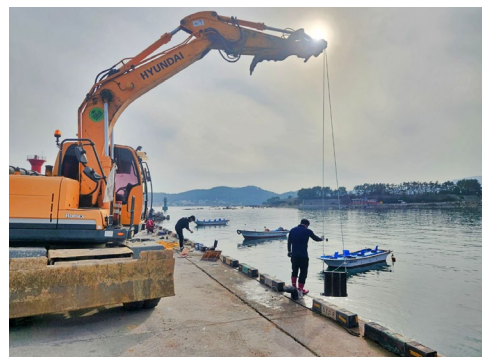
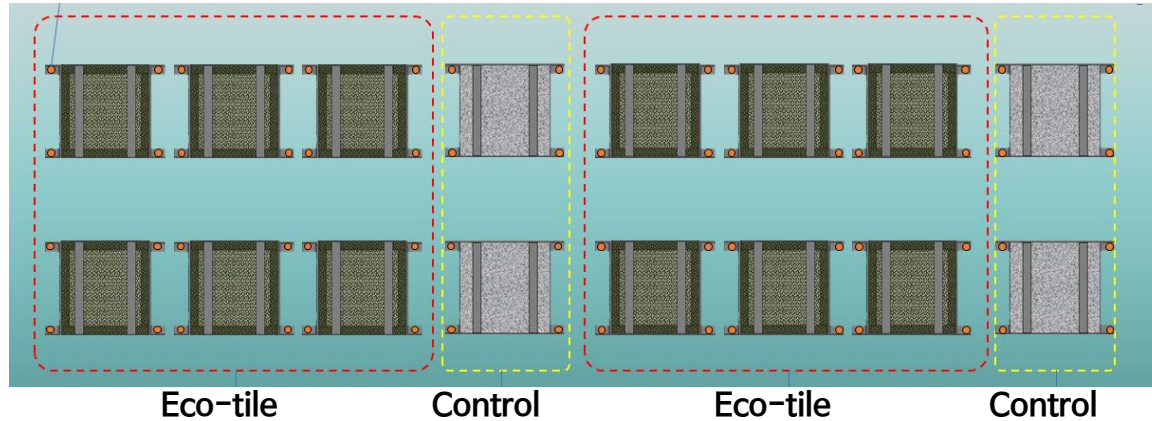


3. Key Findings & Discussion

BC criteria for Tidal flats: Practical management to carbon reduction

④ Soft Living Test-bed 1

(Location: Busan; Monitoring: Nov. 2022 ~ Nov. 2024 24 month)

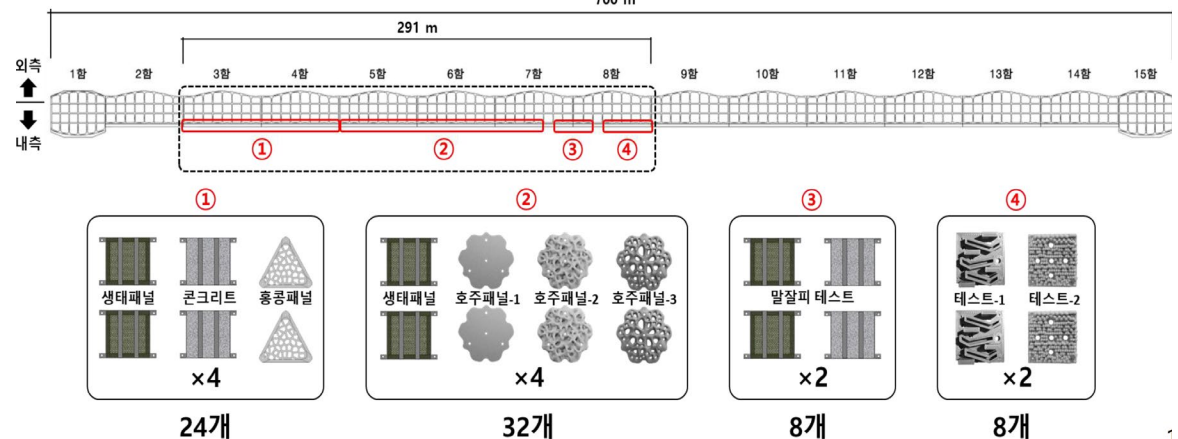


⑤ Soft Living Test-bed 2

(Location: Pohang; Monitoring: Nov. 2023 ~ Dec. 2026, 38 month)



❖ 소프트리빙 테스트베드 조성 규모
- 연성화 구조물 총 72개 설치

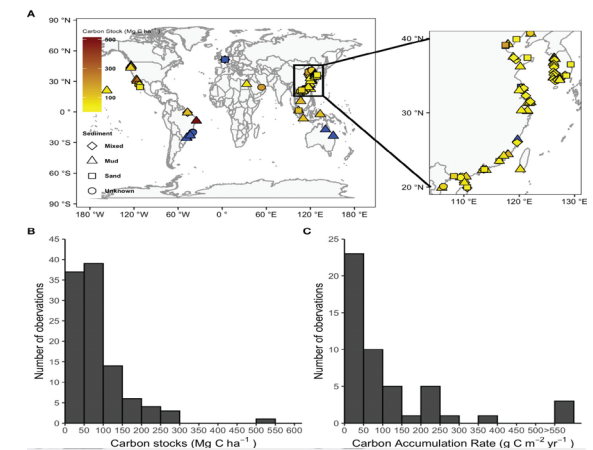
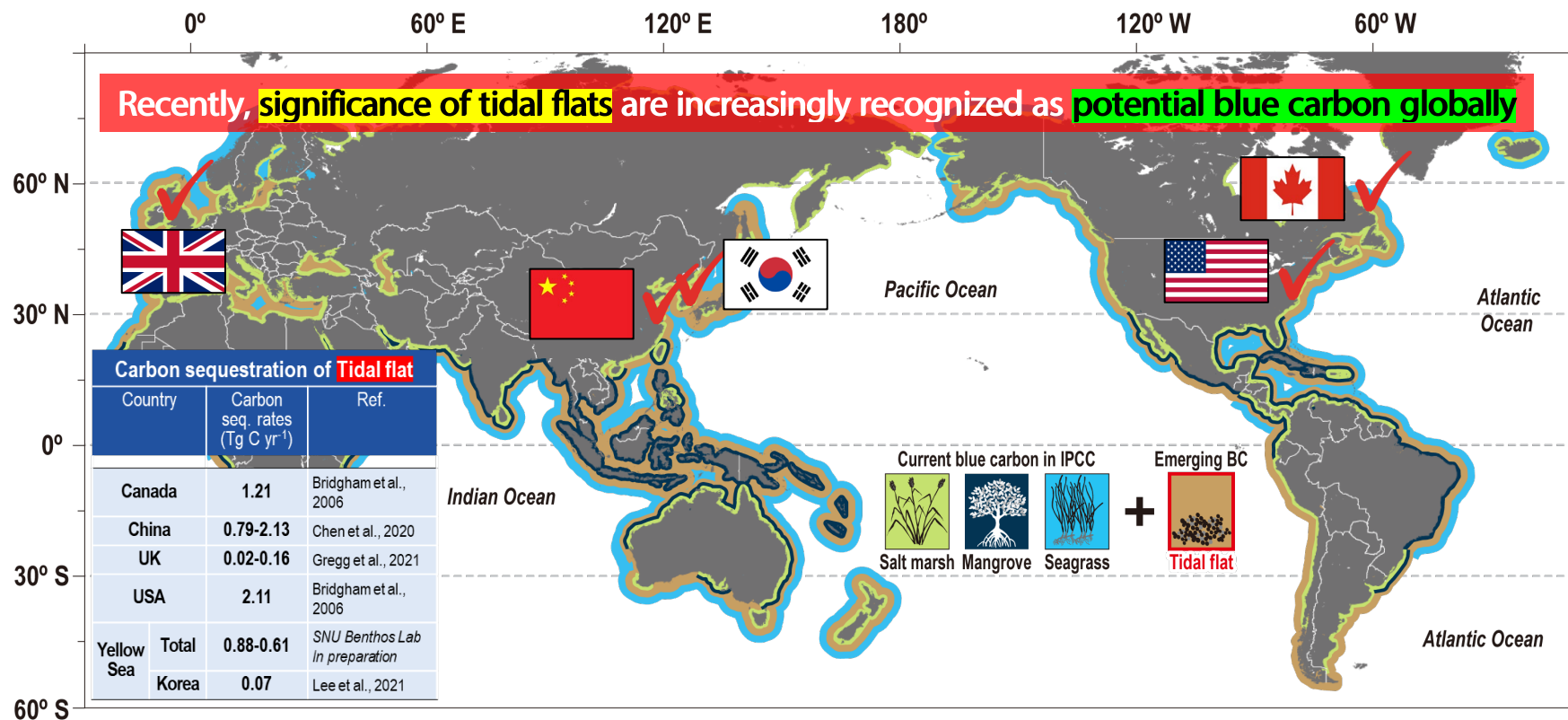


3. Key Findings & Discussion

BC criteria for **Tidal flats**: Included in IPCC GHG accounting guidelines

Blue Carbon Criteria (Adapted from Lovelock & Duarte, 2019; Pidgeon et al., 2021)

Category	Habitat type	Large scale of GHG removals or emissions	Long-term storage of fixed CO ₂	Carbon loss by anthropogenic impacts	Practical management to reduce emission	Included in IPCC GHG accounting guidelines	Alignment with other policies
Emerging Blue Carbon Ecosystems	Tidal flat	? → YES	? → YES	YES	? → YES	NO → YES	YES



Tidal Flats as a Significant Carbon Reservoir in Global Coastal Ecosystems

Tidal flats are widely distributed and provide a variety of ecosystem services. Nevertheless, the consequences of tidal flat loss and implications for services such as carbon (C) sequestration have...

Front. Mar. Sci., 30 May 2022

3. Key Findings & Discussion

2030 NDC & 2050 Carbon Neutrality Scenario: Marine, South Korea

국정현안관계장관회의

블루카본 추진전략

K-Blue Carbon National Strategy

2023. 5. 31.

관계부처 합동

Target	Subject	Unit	GHGs Reduction Target		
			2017 (Base year)	2030 (NDC)	2050 (Carbon Neutrality)
Salt marsh & Seagrass 	Salt marsh planting (Salt marsh, Actionable Blue Carbon)	km ²	0	105	660
		tCO ₂ eq	0	36,471	229,249
	Seagrass planting (Seagrass, Actionable Blue Carbon)	km ²	0	1	3
		tCO ₂ eq	0	164	493
Tidal flat 	Tidal flat restoration (Tidal flat, Emerging Blue Carbon)	km ²	0	10	30
		tCO ₂ eq	0	1,980	5,940
	Inshore protected area (Tidal flat, Emerging Blue Carbon)	km ²	0	1,318	1,418
		tCO ₂ eq	0	260,964	280,764
	Offshore protected area (Benthic sediment, Emerging Blue Carbon)	km ²	0	9,253	9,253
		tCO ₂ eq	0	577,387	577,387
Marine forest 	Marine forest & artificial reef (Macroalgae, Emerging Blue Carbon)	km ²	0	388	388
		tCO ₂ eq	0	130,885	30,885
	Island sea foresting (Macroalgae, Emerging Blue Carbon)	km ²	0	30	110
		tCO ₂ eq	0	10,120	36,938
	Sea shelf restoration (Macroalgae, Emerging Blue Carbon)	km ²	0	10	34
		tCO ₂ eq	0	3,373	11,301
Oyster shell	Oyster culture & recycling (Oyster, Not Actionable Blue Carbon)	km ²	0	102,000	204,000
		tCO ₂ eq	0	44,880	89,760
Total		tCO ₂ eq	0	1,066,226	1,362,717

Published Works since 2000s

21 / 23

4. Remarks

Collaboration Efforts since 2016





Thank you for listening!



Ministry of Oceans
And Fisheries



Blue Carbon
Research Center